Goal

In this assignment I implemented the Canny Edge detector which uses multiple steps in order to get a very thorough edge detection for images. Alongside this, there are several parameter and options which have been tested to fine tune the performance

Algorithm

Gaussian Smoothing

A gaussian matrix is convolved with the image to smooth out features in the image. The parameter N and sigma are tunable to improve the performance

Image Gradient

In this step, each pixel's gradient and direction is determined using a designed matrix. I explored the sobel and roberts matrix in my exploration

Thresholds

The histogram of the magnitude matrix is calculated and another tunable parameter: percentNonEdge is used to determine strong magnitudes in the photo. Additionally, this T_high is used to determine a T_low that will improve edges in later steps

Surpressing NonMaxima

Only lines which are higher than the values along their normal vector remain after this step. I used the quantization method to determine the normal vector and the pixels to compare against

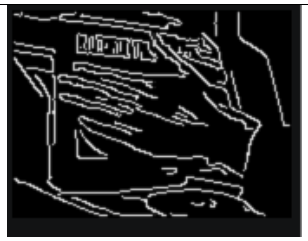
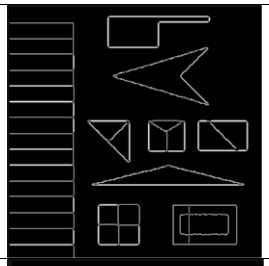
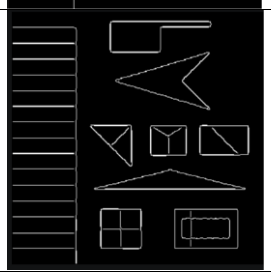
Threshold and Edge Linking

This recursive method analyses every strong edge (> T_high) and connects lines by looking for weak edges (T_high >edges >T_low) that connect the strong edges.

Results

Test Cases

I noticed worse performance with changing sigma but nuanced difference while playing around with N and perfectofNonEdge which balanced capturing details and removing noise. These results were obtained with the **Sobel matrix**

	Lena	Joy	Pointer	Test
N=3 Sigma=1 % = 0.8				
N=5 Sigma=1 % = 0.75				

The following results were with the **Roberts Cross Matrix.** I believe further tuning would be needed with this matrix since I was noticing gaps among the edges across the joy and pointer photo specifically. I did, however, notice really great capture of the edges on the Lena image

